British Math Olympiad 1997 Rd 1 — P4/5

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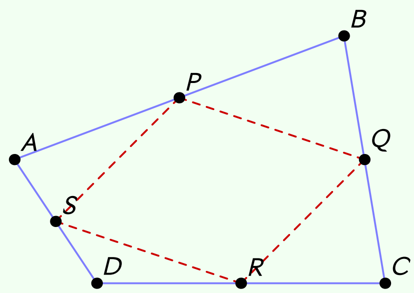
Let $ABCD$ be a convex quadrilateral. The midpoints of AB, BC, CD and DA are P, Q, R, and S, respectively. Given that the quadrilateral $PQRS$ has area 1, prove that the area of the quadrilateral $ABCD$ is 2.

Solution



The idea is to use the fact that $\triangle BPQ \sim \triangle ABC$. This because $\overrightarrow{AB} = 2\overrightarrow{PB}$, $\overrightarrow{CB} = 2\overrightarrow{QB}$ and thus $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = 2(\overrightarrow{PB} + \overrightarrow{BQ}) = 2\overrightarrow{PQ}$. Thus we can conclude $[ABC] : [PBQ] = 4 : 1$. By symmetrical reasoning we have

$$\frac{[PBQ]}{[ABC]} = \frac{[SDR]}{[ADC]} = \frac{[QCR]}{[BCD]} = \frac{[PAS]}{[BAD]} = \frac{1}{4}$$

and so

$$\begin{aligned} [ABCD] &= [ABC] + [ADC] = 4([PBQ] + [ADC]) \\ &= [DAB] + [DCB] = 4([SAP] + [RCQ]) \end{aligned}$$

which gives

$$\begin{aligned} 2[ABCD] &= 4([PBQ] + [ADC] + [SAP] + [RCQ]) \\ [ABCD] &= 2([PBQ] + [ADC] + [SAP] + [RCQ]) \\ &= ([PBQ] + [ADC] + [SAP] + [RCQ]) + [PQRS] \end{aligned}$$

and so we can conclude that $([PBQ] + [ADC] + [SAP] + [RCQ]) = [PQRS] = 1$. Now substituting gives that $[ABCD] = 2$. ■